CIVICSIFT AI

Topic Modeling and Extractive Text Summarization

Natural Language Processing Project Report

WEEK 5

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Abstract

This report presents the Week 5 development of CivicSift AI, an NLP-based project that explores common themes in NYC 311 complaint descriptions and produces concise, representative summaries. The workflow includes text preprocessing, Latent Dirichlet Allocation (LDA) topic modeling, TF-IDF-based extractive summarization and quantitative evaluation. The source dataset contained 369 records; 366 records were processed successfully. An LDA model was trained with four topics using a document-term matrix containing 195 vocabulary features. Representative complaint descriptions were selected for each topic using TF-IDF similarity. The evaluation script reported a perplexity of 293.686213, topic diversity of 1.000000, average topic-assignment confidence of 0.611134 and average summary relevance of 0.318100. The results demonstrate a complete working pipeline while also revealing limitations caused by short complaint descriptions, a relatively small dataset and imperfect topic coherence.

1. Introduction

Civic complaints often contain recurring issues relating to public infrastructure, sanitation, noise, traffic and municipal services. Manually reviewing large collections of complaints can make it difficult to identify repeated themes quickly. CivicSift AI explores how NLP techniques can organize complaint text into topics and select representative examples that help communicate the main content of each topic.

The Week 5 implementation focuses on unsupervised topic discovery and extractive summarization. It does not generate new natural-language sentences; instead, it selects existing complaint descriptions that are representative of each discovered topic.

2. Objectives

- Clean and prepare complaint descriptions for NLP processing.
- Convert the cleaned text into numerical document-term features.
- Apply LDA to discover four initial topics in the complaint dataset.
- Assign each complaint to its dominant topic.
- Generate extractive summaries by selecting representative complaints using TF-IDF similarity.
- Evaluate topic modeling and summary relevance using the implemented metrics.

3. Dataset Description

The implementation used the file nyc_311_complaints.csv. The preprocessing output indicates 369 original records, of which 366 were processed. Three records were removed during preprocessing. The cleaned output contained zero empty cleaned records. The scripts expected a complaint-text column named 'text'. The dataset consists of short complaint descriptions and associated complaint category information used when displaying the most common categories in the topic summaries.

Dataset measure	Observed value
Original records	369
Processed records	366
Records removed	3
Empty cleaned records	0
Document-term matrix	366 × 195
Vocabulary features	195

4. Methodology and Implementation

4.1 Text preprocessing

The preprocessing script loads the CSV file with pandas, checks for the required 'text' column, fills missing text values with empty strings, converts the text to string values and applies lowercasing and regular-expression cleaning. English stop words are removed using scikit-learn's ENGLISH_STOP_WORDS. The processed dataset is saved as outputs/preprocessed_complaints.csv.

4.2 LDA topic modeling

The topic-modeling script reads the cleaned complaint text and converts it into a document-term matrix using CountVectorizer. The configured vectorizer uses max_features=1000, min_df=2, max_df=0.95 and ngram_range=(1, 2). The resulting matrix had 366 documents and 195 vocabulary features. A Latent Dirichlet Allocation model was trained with four topics as the initial experimental setting. Each complaint was assigned a dominant topic and a topic probability. The model, fitted vectorizer, topic keywords and topic-assigned complaints were saved in the outputs folder.

The discovered topic keywords and distribution were:

Topic	Top keywords from the run	Complaints	Share
1	blocked, sidewalk, defective, permit, illegal, work, sign, improper, window, garbage	155	42.35%
2	loud, music, truck, construction, door, music party, loud music, party, loud talking, talking	66	18.03%
3	street, missing, street light, light, basin, store, catch basin, catch, line, trash	67	18.31%
4	food, signal, hydrant, odor, veh, non, building, passenger, driver, veh signal	78	21.31%

4.3 Extractive text summarization

The summarization script groups complaints by their dominant LDA topic and removes duplicate complaint descriptions within each group. It converts the complaint descriptions into TF-IDF vectors, calculates a topic-level average TF-IDF representation (centroid), and uses cosine similarity to identify descriptions close to that centroid. The selected complaint descriptions are presented as representative complaints. The script also displays the most frequent complaint categories for each topic. The outputs are saved as topic_summaries.csv and topic_summaries.txt.

A NumPy/scikit-learn compatibility issue occurred when the topic centroid was represented as a NumPy matrix. It was corrected by converting the centroid to a standard array with np.asarray(tfidf_matrix.mean(axis=0)). After this correction, the summarization script completed successfully.

4.4 Model evaluation

The evaluation script loaded the topic-assigned complaints, topic keywords, generated summaries, trained LDA model and fitted CountVectorizer. It calculated and saved the evaluation metrics reported below. The script completed successfully and generated evaluation_metrics.csv and summary_relevance_by_topic.csv.

5. Summarization Results

Topic	Representative complaints observed	Top complaint categories observed
1	Blocked Sidewalk; Broken Sidewalk; Garbage or Litter; Traffic Sign or Signal Blocked; Sign/Awning/Marquee - Illegal/No Permit	Illegal Parking; General Construction/Plumbing; Sidewalk Condition
2	Lamppost Damaged; No Water; DOOR; DOOR FRAME; DOOR/FRAME	Noise; Water Maintenance; Street Light Condition
3	Street Light Out; Trash or Recycling; Blocking Street; Clear Street Light; Street Light Dayburning	Consumer Complaint; Street Light Condition; Street Condition
4	Non-Food Vendor; Veh Signal Lamp; Fire Hydrant; Food Vendor; Fire Hydrant Running	Traffic Signal Condition; Food Establishment; Water Maintenance

The summaries show that some topics are interpretable, while others contain mixed complaint types. For example, Topic 3 contains several street-light and street-maintenance descriptions. Topic 2's keywords suggest noise-related complaints, but its selected representative descriptions include door and water-related entries. This indicates that the topic grouping is not consistently coherent.

6. Evaluation Results

Metric	Observed score
LDA Perplexity	293.686213
Topic Diversity	1.000000
Average Topic Assignment Confidence	0.611134
Average Summary Relevance	0.318100

6.1 Summary relevance by topic

Topic	Summary relevance
Topic 1	0.2660
Topic 2	0.3278

Topic 3	0.3468
Topic 4	0.3318

Topic 3 obtained the highest reported summary relevance score (0.3468), while Topic 1 had the lowest (0.2660). The overall average summary relevance was 0.318100. The average topic-assignment confidence was 0.611134. These values describe the outputs of this implementation; they should not be interpreted as classification accuracy or proof of semantic correctness.

The reported topic diversity score was 1.000000. This is the value produced by the implemented evaluation script. Its interpretation depends on the exact calculation used; a high diversity score alone does not establish that topics are semantically coherent. LDA perplexity is most useful for comparison between models evaluated on the same data with consistent preprocessing and vectorization settings. The value 293.686213 should therefore be treated as a baseline result, not as a standalone quality verdict.

7. Discussion

The Week 5 pipeline successfully completed the main stages from raw complaint text to topic-based representative summaries and evaluation outputs. The four-topic model produced distinguishable keyword groups and assigned all 366 processed complaints to dominant topics. Topic 1 contained the largest share of complaints (42.35%).

However, the topics are not perfectly separated. Some keywords and representative complaint descriptions do not align cleanly. This is particularly visible in Topic 2 and Topic 4, where different complaint types appear together. The short length of the complaint descriptions provides limited context for discovering stable semantic themes. The dataset size and the initial choice of four topics also limit the strength of conclusions that can be drawn.

8. Challenges and Limitations

- The dataset is relatively small for discovering robust topic structures.
- Complaint descriptions are often short phrases rather than detailed documents, so they provide limited contextual information.
- Some discovered topics contain mixed or overlapping complaint types.
- The four-topic setting was used as an initial experiment; the current run does not by itself prove that four is the optimal number of topics.
- Extractive summarization selects existing complaint descriptions and does not create fluent, newly generated summaries.
- No human-written reference summaries were available, so ROUGE-based evaluation was not reported.
- Perplexity and topic diversity should be compared across consistent experimental settings and interpreted alongside manual topic inspection.

9. Future Improvements

- Evaluate multiple topic counts using the same preprocessing and vectorization settings, then compare perplexity, topic diversity and human interpretability.
- Use a larger and more varied complaint dataset to improve topic discovery.
- Retain more contextual fields, where available and appropriate, to support richer analysis.
- Review topic keywords and representative complaints manually to assess topic coherence.
- Explore alternative topic modeling approaches and compare their outputs with the LDA baseline.
- Create human-reviewed reference summaries if a ROUGE-based summarization evaluation is required.

10. Conclusion

Week 5 of CivicSift AI implemented a complete experimental workflow for topic-based analysis and extractive summarization of NYC 311 complaints. Text preprocessing produced 366 usable records from 369 original records. LDA discovered four initial topics from a 195-feature document-term matrix, and TF-IDF with cosine similarity was used to select representative complaints for each topic. The evaluation script reported a perplexity of 293.686213, topic diversity of 1.000000, average topic-assignment confidence of 0.611134 and average summary relevance of 0.318100. Although the implementation ran successfully, the mixed content in some topics and the short complaint descriptions show that further experimentation and a larger dataset are needed. Overall, this week demonstrates the practical use of LDA and extractive summarization in organizing civic complaint data.

11. Generated Project Outputs

- outputs/preprocessed_complaints.csv
- outputs/lda_topics.csv
- outputs/complaints_with_topics.csv
- outputs/lda_model.pkl
- outputs/count_vectorizer.pkl
- outputs/topic_summaries.csv
- outputs/topic_summaries.txt
- outputs/evaluation_metrics.csv
- outputs/summary_relevance_by_topic.csv

12. Reproducibility Notes

The project was implemented in Python in PyCharm. The scripts were organized as 01_preprocessing.py, 02_topic_modeling.py, 03_text_summarization.py and 04_evaluation.py. The dataset was placed in the data folder and generated files were stored in outputs. The reported numerical results are transcribed from the successful execution output supplied for this project. For reproducibility, retain the source dataset, requirements.txt and all four scripts alongside the generated outputs.